System Test Report

1. Introduction

System testing is performed to verify that the complete Hostel Management System works as expected when its different modules interact with one another.

The purpose of this testing is to validate the application's major workflows, including student authentication, room management, fee payments, leave applications, mess management, complaints and warden operations.

The system was tested using automated HTTP requests through the Flask Test Client, along with database operations using a separate testing database.

2. Objectives of System Testing

- To verify the overall functionality of the Hostel Management System.
- To ensure that different application modules work together correctly.
- To validate complete user workflows.
- To verify communication between the application and database.
- To check authentication and role-based access control.
- To verify that invalid operations and unauthorized requests are handled correctly.

3. Testing Environment

Component	Description
Application	Hostel Management System
Programming Language	Python
Web Framework	Flask
Testing Framework	Pytest
Database	SQLite (Testing Database)
ORM	SQLAlchemy
Testing Interface	Flask Test Client
Operating System	macOS, Windows
Execution Environment	Python Virtual Environment

The testing environment used a separate SQLite database to avoid modifying the application's actual database records.

4. System Test Scenarios and Test Cases

4.1 Student Authentication and Dashboard

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-001	Open the student login page	Login page should be displayed	Login page loaded successfully	PASS
ST-002	Log in using valid student credentials	Student should be redirected to the dashboard	Student was redirected to the dashboard	PASS
ST-	Log in using incorrect	System should reject the login	Login was rejected	PASS

003	credentials			
ST-004	Access the dashboard after login	Dashboard should display student information and available modules	Dashboard loaded successfully	PASS
ST-005	Log out from the application	Student session should be terminated	Session was terminated	PASS
ST-006	Access a protected page without authentication	System should redirect the user to login	User was redirected	PASS

4.2 Room Management and Availability

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-007	Open the room details page	Student room details should be displayed	Room details page loaded	PASS
ST-008	View room availability	Available, full and maintenance room statuses should be displayed	Room statuses were displayed	PASS
ST-009	Filter rooms by hostel	Only rooms belonging to the selected hostel should be displayed	Filtering worked as expected	PASS
ST-010	Access room allocation without login	Access should be restricted	User was redirected to login	PASS
ST-011	Attempt an unsupported room allocation operation	System should prevent the operation	Allocation operation was rejected	PASS

4.3 Fee and Payment Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-012	View student fee details	Student should be able to view their fee records	Fee details were displayed	PASS
ST-013	Pay an existing fee	Payment should be recorded and fee status updated	Payment was recorded successfully	PASS
ST-014	Generate a payment receipt	Receipt should be generated and displayed	Receipt was displayed	PASS
ST-015	Attempt to pay another student's fee	System should prevent unauthorized payment	Request was rejected	PASS
ST-016	Warden views student fee records	Warden should be able to view fee records	Fee records were displayed	PASS

4.4 Leave Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-017	Submit a valid leave application	Application should be saved	Application was saved	PASS
ST-018	View submitted leave applications	Student should be able to view their applications	Applications were displayed	PASS
ST-019	Submit an invalid leave application	System should reject invalid application details	Invalid application was rejected	PASS
ST-020	Attempt to modify another student's leave application	System should prevent unauthorized modification	Unauthorized operation was rejected	PASS

4.5 Common Facility Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-021	View available common facilities	Facility details should be displayed	Facility details were displayed	PASS
ST-022	Book an available facility	Booking should be recorded	Booking was recorded	PASS
ST-023	Submit an invalid facility booking	System should reject invalid booking details	Invalid booking was rejected	PASS
ST-024	Attempt to book an unavailable facility	System should prevent the booking	Booking was rejected	PASS

4.6 Mess Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-025	View the weekly mess menu	Weekly menu should be displayed	Menu was displayed	PASS
ST-026	Register for a meal	Meal registration should be recorded	Registration was recorded	PASS
ST-027	View mess attendance	Attendance information should be displayed	Attendance information was displayed	PASS
ST-028	Submit mess feedback	Feedback should be saved	Feedback was saved	PASS
ST-029	View mess fee information	Mess fee details should be displayed	Fee information was displayed	PASS
ST-030	View mess holidays	Holiday information should be displayed	Holiday information was displayed	PASS

4.7 Complaints and Lost and Found

Test ID	Test Scenario	Expected Result	Actual Result	Status
ST-031	Open the complaints page	Complaints page should load	Page loaded successfully	PASS
ST-032	Submit a valid complaint	Complaint should be recorded	Complaint was recorded	PASS
ST-033	View submitted complaints	Student should be able to view complaint details and status	Complaint details were displayed	PASS
ST-034	Submit a complaint with an attachment	Attachment should be saved successfully	Attachment was saved	PASS
ST-035	View lost and found items	Item details should be displayed	Item details were displayed	PASS
ST-036	Submit a found item	Item should be recorded	Item was recorded	PASS

4.8 Warden Operations and Access Control

Test	Test Scenario	Expected Result	Actual Result	Status

ID				
ST-037	Log in using valid warden credentials	Warden should be redirected to the warden dashboard	Warden was redirected successfully	PASS
ST-038	Access warden dashboard as a student	Access should be denied	Access was denied	PASS
ST-039	Access warden functionality without authentication	User should be redirected to login	User was redirected	PASS
ST-040	Create a mess menu entry as a warden	Menu entry should be saved	Menu entry was saved	PASS
ST-041	Update an existing mess menu entry	Menu details should be updated	Menu was updated	PASS
ST-042	Attempt to create a duplicate menu entry	System should reject duplicate entries	Duplicate entry was rejected	PASS
ST-043	Attempt to delete a menu entry with registrations	System should prevent deletion	Deletion was prevented	PASS

Note: These are representative scenarios derived from the automated test suite. They are not a claim that all 43 scenarios were separately executed as manual end-to-end tests. The actual automated tests cover these and additional cases.

5. Test Execution Summary

The complete automated test suite was executed using Pytest.

Parameter	Result
Total Automated Tests	223
Tests Passed	223
Tests Failed	0
Warnings	7
Overall Result	PASS

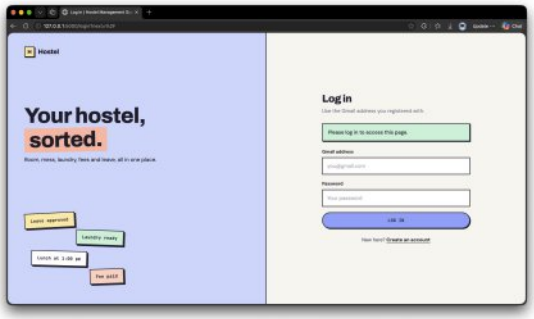
The automated test suite completed successfully, with all 223 tests passing.

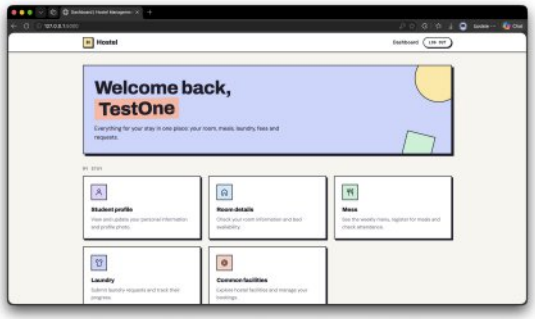
The successful GitHub Actions workflow also confirms that the pushed test suite completed successfully in the CI environment.

6. Testing Evidence

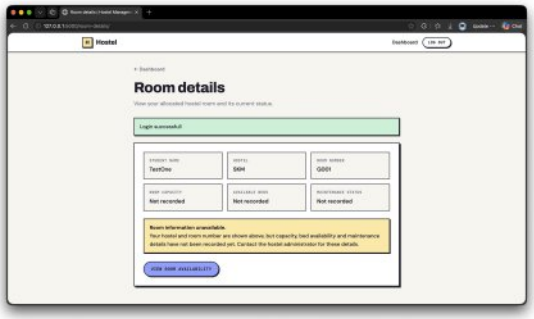
The following screenshots can be included as evidence:

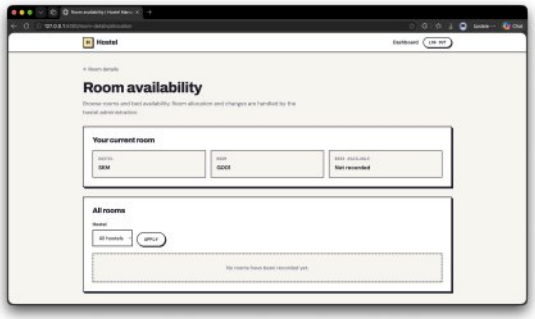
1. Screenshot of the successful student login and dashboard.



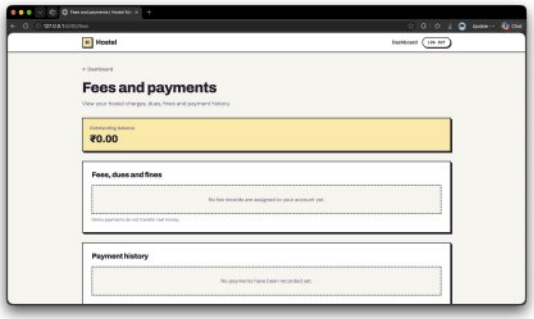


2. Screenshot of the room details and room availability pages.

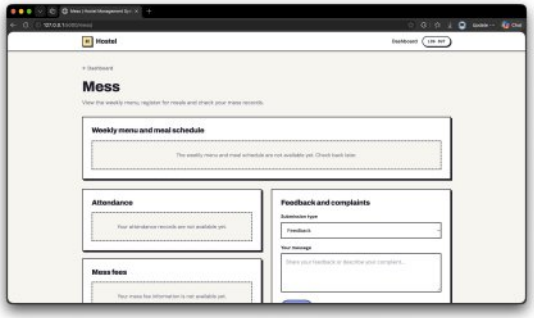




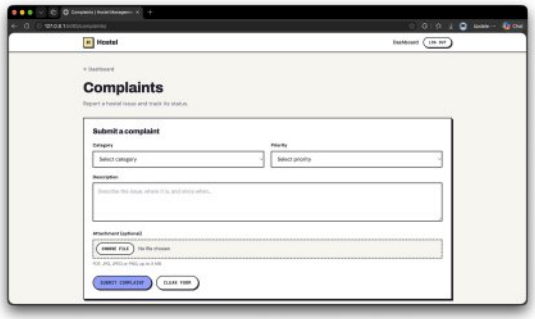
3. Screenshot of the fee payment and receipt pages.



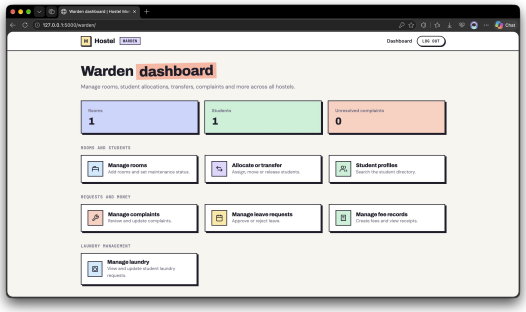
4. Screenshot of the mess management page.



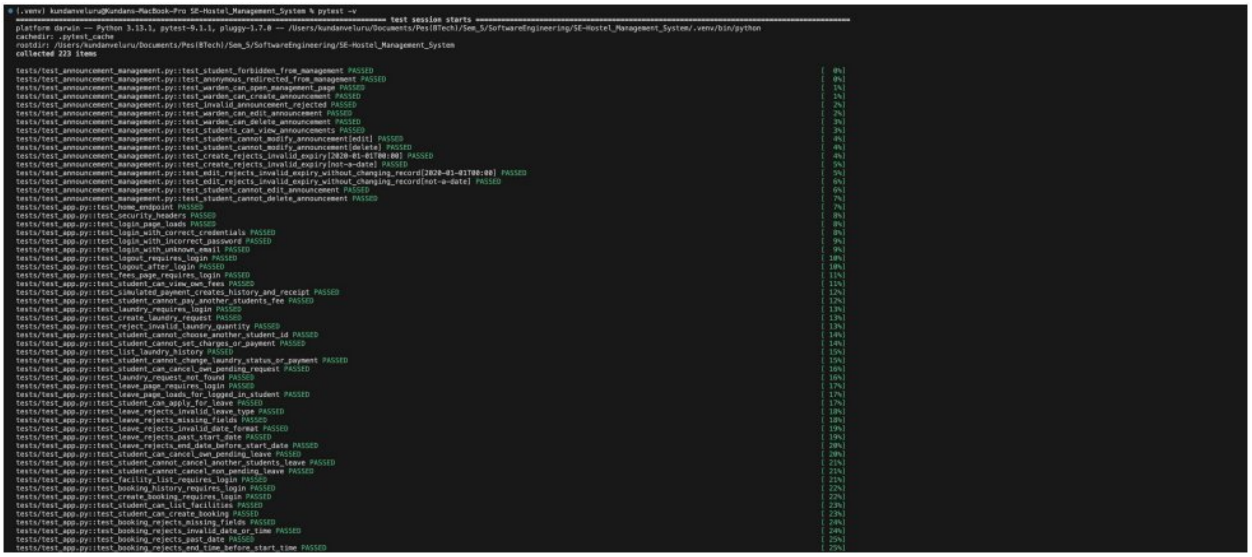
5. Screenshot of the complaints page.



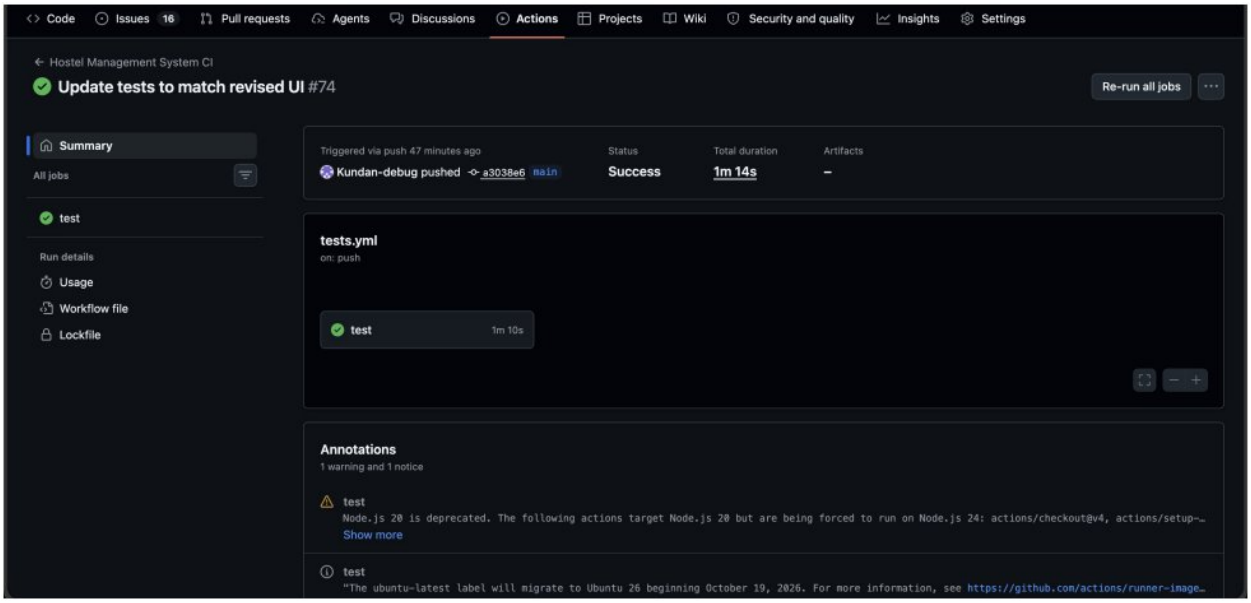
6. Screenshot of the warden dashboard.



7. Screenshot of the terminal showing the successful Pytest execution.



8. Screenshot of the successful GitHub Actions workflow.



Automated Testing Command:

pytest -v

7. Conclusion

System testing was performed on the Hostel Management System to verify the interaction between its major application modules.

The testing covered student authentication, dashboard access, room availability, fee payments, leave applications, common facility booking, mess management, complaints, lost and found, and warden operations.

The automated test suite completed with 223 passed tests, zero failed tests and seven warnings. The GitHub Actions workflow also completed successfully.

The results demonstrate that the tested application workflows and interactions behaved as expected under the automated testing conditions.